Md Sifatullah Sheikh

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RESEARCH INTERESTS

Trustworthy and explainable AI, computer vision, multimodal learning, and the safety of AI systems that see and act (vision-language-action models). My work so far asks one question: when should we trust what a vision model tells us, whether it is judging a face as real or naming a medicinal plant?

EDUCATION

B.Sc. in Computer Science and Engineering Jan 2022 – Feb 2026
East West University, Dhaka, Bangladesh
CGPA 3.70/4.00 · Major: Data Science and Intelligent Systems · Dean's List
Thesis: *MediLeafNET: Multimodal Medicinal Plant Identification and Knowledge Retrieval*
Supervisor: Dr. Md. Nawab Yousuf Ali

PUBLICATIONS

Google Scholar: 20 citations.

[J1] Md Al-Imran, **Md Sifatullah Sheikh**, Urmi Kirtonia, Nuzath Tabassum Arthi, Shamim Ripon. "DeFaX: A Cross-Attention Fusion Framework for Robust and Explainable Deepfake Detection." *IEEE Access*, vol. 13, pp. 213962–213979, 2025. doi:10.1109/ACCESS.2025.3645769

[C1] **Md Sifatullah Sheikh**, Urmi Kirtonia, Nuzath Tabassum Arthi, Md Al-Imran. "AI-Powered Deepfake Detection Using CNN and Vision Transformer Architectures." *6th International Conference on Big Data Analytics and Practices (IBDAP)*, IEEE, Thailand, 2025. Oral presentation. IEEE Xplore

[D1] "A Medicinal Plant Leaf Image Dataset of 13 Species Collected in Bangladesh." *Mendeley Data*, Version 2. Contributed to image collection and public release. doi:10.17632/9tdc9gbtgb.2

RESEARCH EXPERIENCE

Research Assistant, AI-Generated Content Detection Dec 2024 – Dec 2025
Department of CSE, East West University · with Md Al-Imran and Dr. Shamim H Ripon

- Benchmarked four deep learning architectures for detecting GAN- and AI-generated faces on 140,000 images, and found that CNNs and transformers fail on different cases [C1].
- Designed DeFaX, a cross-attention fusion of Swin Transformer (global context) and EfficientNet (local texture); scaled the study to about 250,000 images with cross-dataset tests [J1].
- Ran ablation studies, layer freezing experiments and Grad-CAM analysis to check that the model looks at manipulated regions, not dataset shortcuts.

Undergraduate Thesis, MediLeafNET Feb 2025 – Dec 2025
East West University · code · live demo

- Built a system that identifies a medicinal plant from a leaf photo and answers questions about its uses, compounds and safety, combining a vision model with a language model grounded in a curated knowledge base.
- Collected 3,289 leaf images of 11 species at the National Botanical Garden, Bangladesh, and released the extended 13-species dataset publicly [D1].
- Compared several deep learning models on an independent benchmark (best: 94.45% accuracy) and used visual explanations to check which leaf regions drive each prediction.

PROFESSIONAL EXPERIENCE

Trainee Software Engineer (AI/ML) Aug 2026 – Present
Nasir Syntax Solution Limited (Nasir Group), Dhaka

- Work on depth estimation for Nasir electric bikes and test machine learning models on real-world data.

IT Intern Feb 2026 – May 2026
Banglalink Digital Communications Ltd., Dhaka

- Tested IoT devices and networks, monitored sensor data, and helped set up and support IoT solutions; maintained 100+ operational records and reports.

TEACHING AND MENTORING

Undergraduate Student Mentor 2024 – 2025
Department of CSE, East West University

- Ran voluntary study sessions on data structures, C++ and object-oriented programming for 40+ junior students.

Instructor Jan 2023 – Jul 2024
Fahim's Tutorial Centre, Dhaka

- Taught 100+ school and college students; prepared and graded tests and gave each student individual feedback.

SELECTED PROJECTS

SkinCare AI: Skin Lesion Detection with an LLM Advisor May 2026
github.com/SifatSwapnil2022/SkinCareAI

- End-to-end pipeline from lesion detection (YOLOv8, 95.42% mAP) to an ensemble classifier (ResNet50, EfficientNetB0, MobileNetV2) to LLM-generated advice, served with FastAPI and Docker.

HONORS AND AWARDS

Finalist, ICT Olympiad Bangladesh, Season 3 (national competition) 2026
Dean's List and Dean's Scholarship, East West University (three consecutive semesters) 2023 – 2025
Medha Lalon Merit Scholarship, East West University 2024
3rd Place, IT Olympiad, EWU National RoboFest 2024

SKILLS

Programming	Python, C++, SQL
Machine learning	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV
Methods	Vision Transformers, CNNs, cross-attention fusion, object detection, explainability (Grad-CAM, LIME), ablation studies
Tools	Git, Docker, LaTeX, Google Colab, Kaggle, FastAPI, Streamlit, Gradio
Hardware	Raspberry Pi, Arduino, sensor integration
Languages	Bengali (native), English (IELTS 8.0)

SERVICE AND LEADERSHIP

Executive Member, EWU Robotics Club: organised technical events and workshops 2023 – 2025
Volunteer, LMH Foundation: food and clothing drives for families in need 2015 – Present

REFERENCES

Dr. Shamim H Ripon
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Co-author [J1]

Dr. Md. Nawab Yousuf Ali
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Thesis supervisor

Md Al-Imran
Senior Lecturer, Dept. of CSE
East West University
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Research supervisor [J1, C1]